

BANFF, AB, Canada

WMO: 711220

Lat: 51.1936N

Lon: 115.5525W

Elev: 4583

StdP: 12.42

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-21.5	-14.6	-29.1	1.3	-21.1	-22.1	1.9	-14.0	16.2	19.6	14.7	20.6	2.6	220	0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.5	81.8	56.2	78.4	55.3	74.7	54.3	58.7	74.7	57.1	72.9	55.6	70.9	7.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.7	72.7	61.4	51.5	67.0	59.6	49.6	62.3	58.4	27.5	74.5	26.3	73.1	25.3	71.3	66.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	12.4	11.4	DB	-33.0	86.7	7.9	3.5	-38.6	89.2	-43.2	91.2	-47.6	93.2	-53.3	95.7	(4)
(5)			WB	-32.5	61.8	6.9	1.9	-37.5	63.2	-41.5	64.3	-45.4	65.4	-50.4	66.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.9	17.1	19.7	27.8	36.6	44.9	51.8	58.5	57.0	48.9	38.1	25.4	15.6		
	DBStd	17.66	13.95	12.54	11.34	7.41	6.81	5.54	5.62	5.71	7.19	7.84	11.16	12.85		
	HDD50	5486	1021	848	687	404	187	42	6	11	106	372	737	1065		
	HDD65	10272	1486	1268	1152	852	623	395	210	252	484	833	1187	1530		
	CDD50	701	0	0	0	1	29	97	271	227	72	4	0	0		
	CDD65	15	0	0	0	0	0	1	10	4	0	0	0	0		
	CDH74	890	0	0	0	0	7	62	423	333	65	0	0	0		
	CDH80	170	0	0	0	0	0	10	95	60	5	0	0	0		
	WSAvg	5.2	6.4	5.5	5.4	5.0	4.8	4.5	4.7	4.3	4.6	5.4	6.0	6.2		
	PrecAvg	19.2	1.3	0.9	0.9	1.5	2.4	2.4	2.0	2.1	1.9	1.2	1.2	1.4		
(15) Precipitation	PrecMax	24.3	5.5	4.3	2.0	2.9	6.0	4.2	4.5	5.9	3.3	3.5	2.4	3.9		
	PrecMin	13.9	0.0	0.2	0.3	0.4	0.7	0.7	0.4	0.3	0.3	0.3	0.1	0.0		
	PrecStd	2.9	1.1	0.9	0.5	0.6	1.3	0.9	1.1	1.1	0.8	0.7	0.7	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.9	46.0	55.8	66.4	74.9	80.7	86.7	85.3	80.3	65.9	51.2	42.3		
		MCWB	36.1	35.5	39.7	46.5	51.6	55.6	58.0	57.3	54.5	48.6	41.4	36.1		
	2%	DB	39.4	41.6	49.2	60.2	70.4	74.6	82.9	81.2	75.3	59.3	44.5	37.1		
		MCWB	33.5	33.5	37.5	43.8	49.4	53.7	56.8	55.9	53.4	45.8	37.9	32.2		
	5%	DB	36.3	38.2	45.3	54.7	65.8	70.2	79.0	78.0	70.5	54.8	41.0	33.9		
		MCWB	31.4	31.5	35.5	40.8	47.8	51.7	55.9	55.3	51.8	43.6	35.7	30.3		
	10%	DB	33.2	35.2	41.6	50.0	60.8	66.0	74.9	73.8	64.6	50.3	38.0	31.0		
		MCWB	29.3	29.5	33.6	38.4	45.9	50.3	55.2	54.2	49.9	41.2	33.7	27.7		
	0.4%	WB	37.8	37.2	41.3	47.2	53.4	58.1	61.8	60.3	56.7	50.1	42.1	36.5		
		MCDB	41.8	43.2	52.0	63.6	69.5	73.6	77.4	75.7	73.6	63.4	48.7	40.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.1	34.6	38.6	44.6	50.9	55.4	59.4	58.3	54.5	47.1	39.1	32.9		
		MCDB	38.4	40.2	47.0	58.4	65.6	69.9	75.1	74.5	71.4	57.2	43.2	36.2		
	5%	WB	31.7	32.0	36.4	41.8	48.9	53.3	57.6	56.7	52.5	44.4	36.5	30.5		
		MCDB	35.6	37.6	43.9	52.6	62.9	66.7	73.3	72.8	68.2	53.1	40.3	33.8		
	10%	WB	29.6	29.8	34.1	39.2	46.8	51.5	56.2	55.2	50.4	41.8	34.1	27.9		
		MCDB	33.2	34.8	40.8	48.7	59.6	63.4	71.3	70.5	63.7	49.3	37.5	30.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	20.1	19.3	20.5	22.9	22.4	26.5	27.4	24.2	18.8	14.3	14.5		
		MCDBR	14.0	18.8	22.1	29.7	33.7	32.6	36.2	37.1	36.9	26.4	14.9	12.0		
		MCWBR	9.6	12.4	12.6	15.9	16.3	14.9	14.8	15.4	17.2	14.5	9.7	8.9		
		MCDBR	12.9	17.7	20.6	26.5	29.9	27.8	30.4	31.9	33.5	24.8	13.9	11.9		
	5% WB	MCWBR	9.4	12.0	12.3	14.8	15.0	13.2	13.5	14.2	16.2	14.0	9.5	9.2		
(40) Clear-Sky Solar Irradiance	taub		0.246	0.258	0.285	0.337	0.351	0.369	0.357	0.370	0.328	0.286	0.254	0.236		
	taud		2.370	2.360	2.299	2.168	2.289	2.325	2.375	2.351	2.449	2.549	2.460	2.362		
	Ebn at Noon		255	282	292	285	286	281	282	272	274	270	251	240		
	Edh at Noon		19	25	33	42	39	38	36	35	28	21	17	16		
(44) All-Sky Solar Radiation	RadAvg		350	651	1033	1438	1680	1766	1976	1643	1156	716	379	276		
	RadStd		23	46	70	103	141	146	141	116	134	71	26	17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page